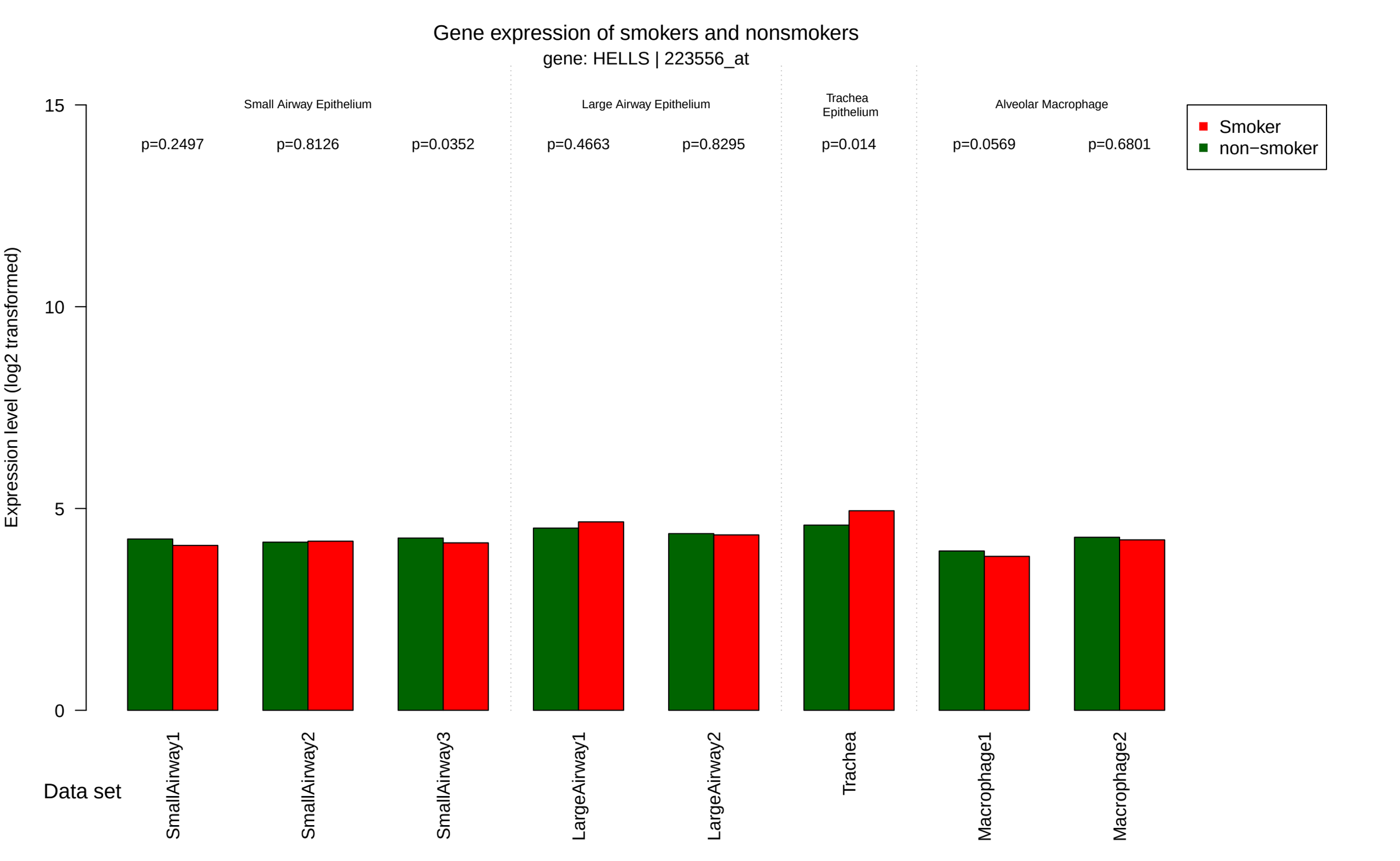
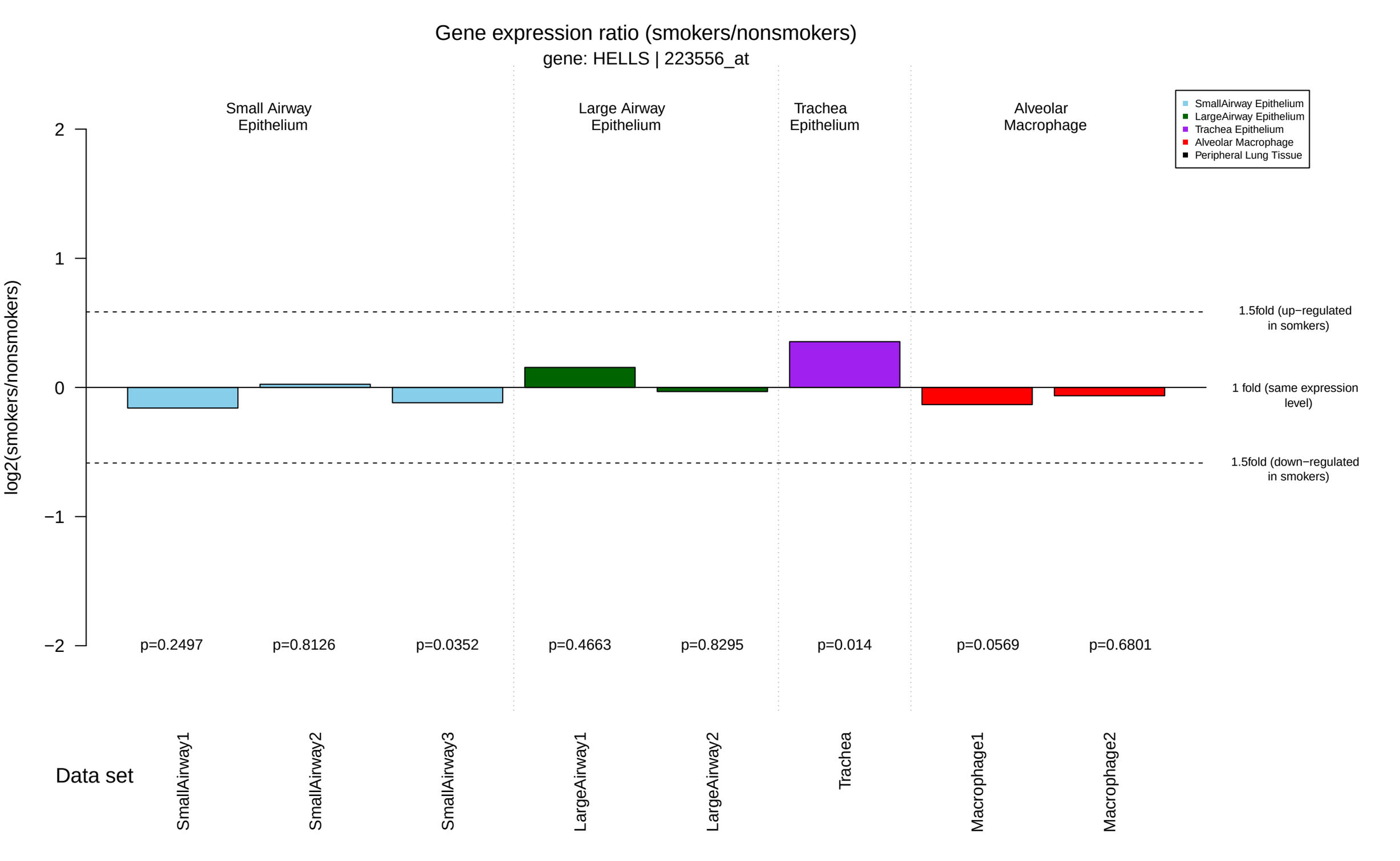
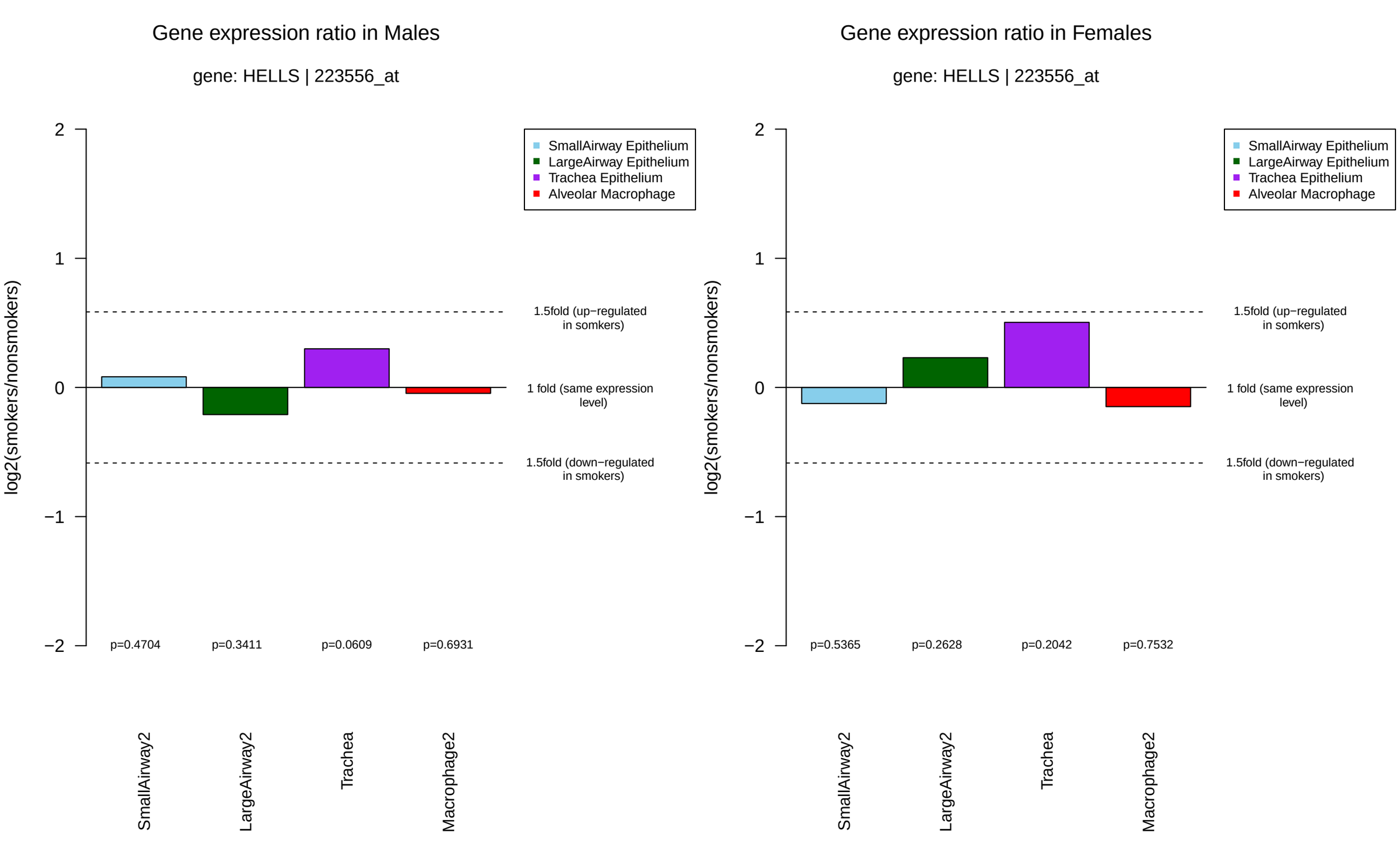
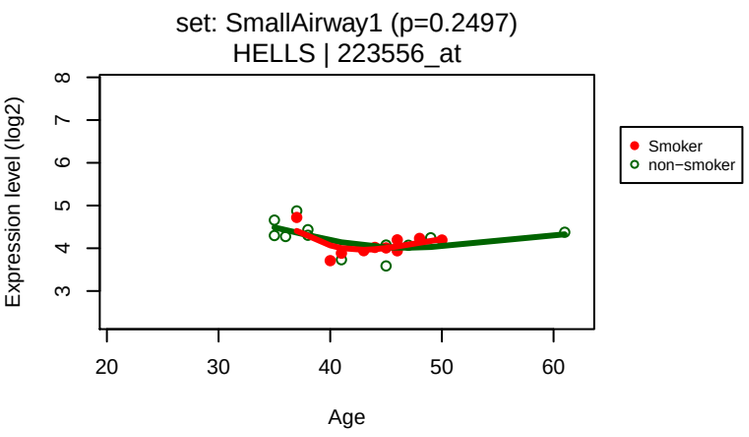
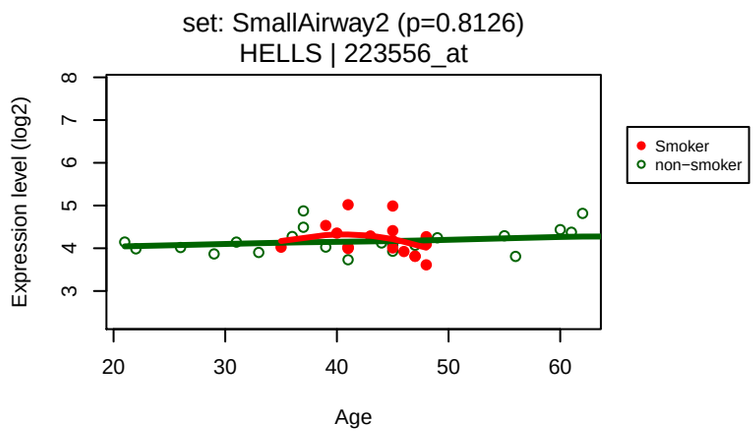
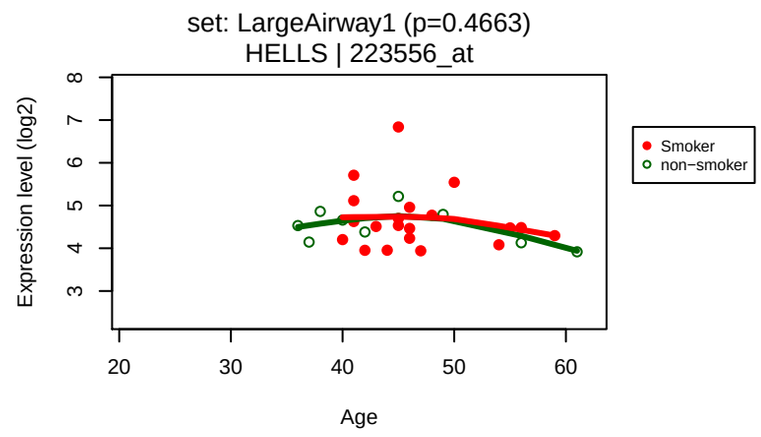

Gene expression of smokers and nonsmokers
gene: HELLS | 223556_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2497
p=0.8126
p=0.0352
p=0.4663
p=0.8295
p=0.014
p=0.0569
p=0.6801
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

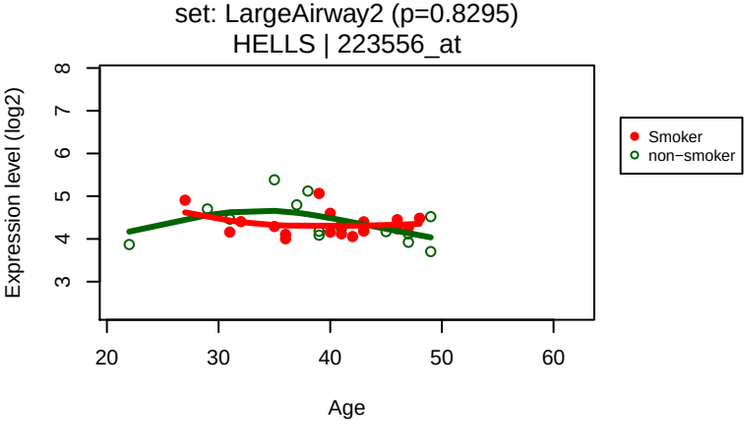

Gene expression ratio (smokers/nonsmokers)
gene: HELLS | 223556_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2497
p=0.8126
p=0.0352
p=0.4663
p=0.8295
p=0.014
p=0.0569
p=0.6801
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

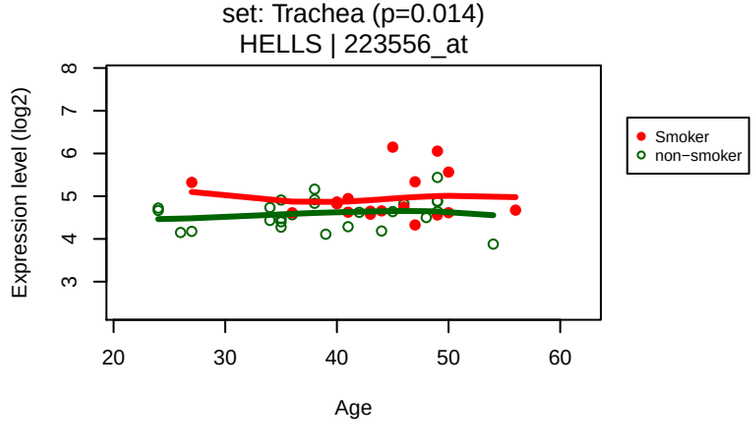

Gene expression ratio in Males
gene: HELLS | 223556_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4704
p=0.3411
p=0.0609
p=0.6931
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HELLS | 223556_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5365
p=0.2628
p=0.2042
p=0.7532
SmallAirway2
LargeAirway2
Trachea
Macrophage2

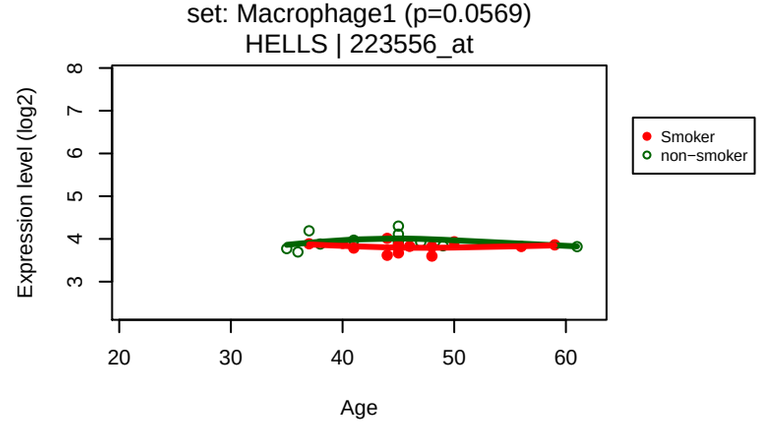

set: SmallAirway1 (p=0.2497)
HELLS | 223556_at
Expression level (log2)
Age
Smoker
non−smoker

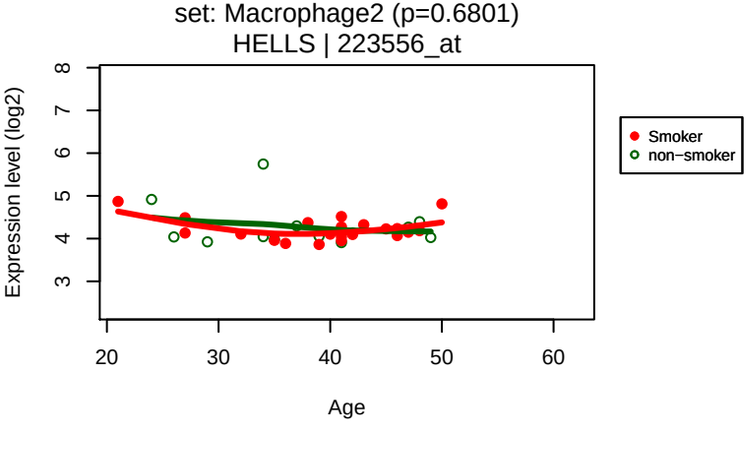

set: SmallAirway2 (p=0.8126)
HELLS | 223556_at
Expression level (log2)
Age
Smoker
non−smoker

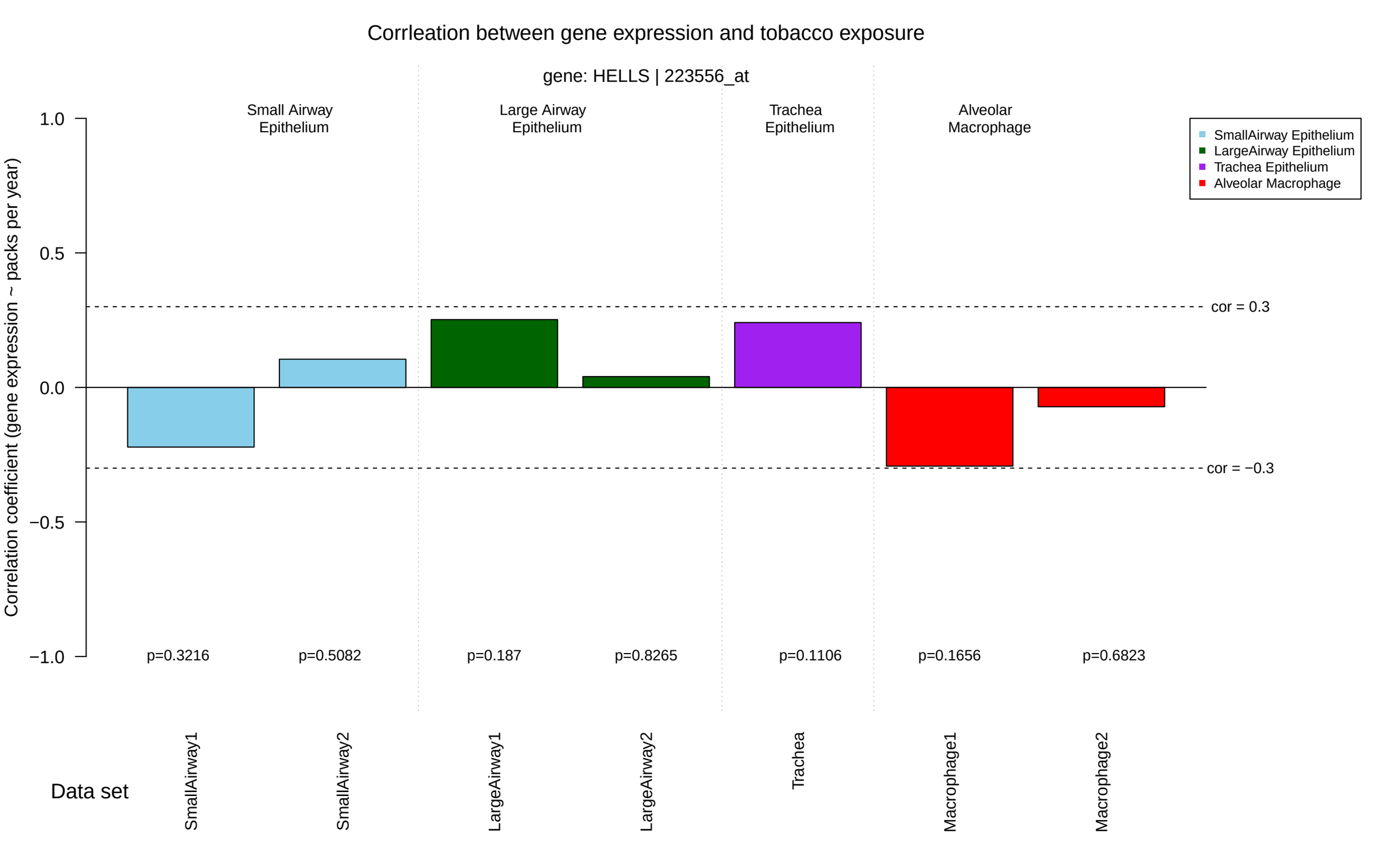

set: LargeAirway1 (p=0.4663)
HELLS | 223556_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8295)
HELLS | 223556_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.014)
HELLS | 223556_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0569)
HELLS | 223556_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6801)
HELLS | 223556_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HELLS | 223556_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3216
p=0.5082
p=0.187
p=0.8265
p=0.1106
p=0.1656
p=0.6823
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2